

N. 1

$$f(x) = \frac{1}{1+2\log(x)} - \sqrt{1+4x-4x^2} \quad 1+4x-4x^2 =$$

Per dobbiamo calcolare il limite per $x \rightarrow 1$ ↓ $\frac{f(x)}{(\sin(\pi x))^2}$

$$\lim_{x \rightarrow 1} \frac{\frac{1}{1+2\log(x)} - \sqrt{1+4x-4x^2}}{(\sin(\pi x))^2} = \frac{\frac{1}{1+2\log(1+\ell)} - \sqrt{1+4(1+\ell)-4(\ell^2+2\ell+1)}}{\sin(\pi(1+\ell))^2} = \frac{1-2\ell+3\ell^2+o(\ell^2) - (1-2\ell-4\ell^2+o(\ell^2))}{\pi^2 \ell^2} =$$

$$\text{Poniamo } \ell = x-1 \rightarrow 0$$

$$= \frac{1-2\ell+3\ell^2+o(\ell^2) - (1-2\ell-4\ell^2+o(\ell^2))}{\pi^2 \ell^2} = \frac{9\ell^2}{\pi^2 \ell^2} = \frac{9}{\pi^2}$$

$$\frac{1}{1+2\log(1+\ell)} = \frac{1}{1+2\ell-\ell^2} = 1-2\ell+\ell^2+\ell\ell^2 = 1-2\ell+3\ell^2+o(\ell^2)$$

$$2\log(1+\ell) = 2\ell - \ell^2$$

$$\sqrt{1-4(\ell^2+\ell)} = 1-2\ell^2-2\ell + \frac{1}{8}(-4\ell^2-4\ell)^2 = 1-2\ell^2-2\ell-2\ell^2 =$$

$$(\sin(\pi(1+\ell)))^2 = \pi^2 \cdot \ell^2 + o(\ell^2)$$

N. 2

$$\tan(x)$$

$$\sin x_0 = \frac{\pi}{4}$$

$$\frac{\tan(x)}{0!} (x-x_0)^0 = 1 + \tan^2(x) = 1$$

$$f''(x) = 2\tan(x)(1+\tan^2(x)) = 4$$

$$f'''(x) = 2(\tan(x) + \tan^3(x))' = 2(1+3\tan^2(x))(1+\tan^2(x)) \Rightarrow f'''(\pi/4) = 16$$

$$\int_0^{\pi/4} \frac{(1-x)(\tan(x))^2}{x^{3/2}(1-\tan(x))^4} dx$$

$$\lim_{x \rightarrow 0^+} \frac{(1-x)(x)^a}{x^{3/2}(1-x)} \sim \frac{1}{x^{3/2-a}} \quad \text{dunque l'integrale converge per } \frac{3}{2}-a < 1 \Rightarrow -a < -\frac{1}{2} \text{ ossia } a > \frac{1}{2}$$

$$\lim_{x \rightarrow \pi/4} \frac{c}{x^{3/2}(1-2(x+\frac{\pi}{4}))^{a-1}} \sim \frac{c_{\frac{\pi}{4}}^{a-1}}{(\frac{\pi}{4}-x)^{a-1}} = \frac{c_{\frac{\pi}{4}}^{a-1}}{(x)^{a-1}} \quad \text{dunque converge per } a-1 < 1 \text{ ossia } a < 2$$

$$\text{dunque l'integrale è convergente} \iff \frac{1}{2} < a < 2$$

U.3

$$\begin{cases} xy'(x) = \frac{2x^2 y(x)}{1-x^2} + 2 \\ y(2) = 3 \end{cases}$$

$$y'(x) - \frac{2x^2 y(x)}{(1-x^2)x} = \frac{2}{x}$$

cerchiamo una primitiva di $q(x) = \frac{-2x^2}{(1-x^2)x}$

$$-2 \int \frac{x}{x^2-1} dx = -\frac{2}{(-2)} \int \frac{2x}{x^2-1} = \ln(x^2-1)$$

Per cui il fattore integrante è

$$e^{\ln(x^2-1)} = (x^2-1)$$

$$y(x) = \int (x^2-1) \cdot \frac{2}{x} dx = 2\ln(x) - x^2 + C$$

$$e^{-A(x)} \int e^{A(x)} \cdot f(x) dx = \frac{1}{(x^2-1)} (2\ln(x) - x^2 + C)$$

Infine imponiamo la condizione

$$y(2) = \frac{1}{(2^2-1)} (2^2 - 2\ln(2) + C) = 3 \Rightarrow \frac{1}{3} (4 - 2\ln(2) + C) = 3 \Rightarrow \frac{1}{3} (4 - 2\ln(2) + 2\ln(2) + 5) = 3$$

$$\frac{4 - 2\ln(2) + C}{3} = 3$$

$$4 - 2\ln(2) + C = 9$$

$$-2\ln(2) = -C + 5$$

$$2\ln(2) = C - 5$$

$$2\ln(2) + 5 = C$$

U.4

$$f(x, y) = xe^y - ye^x$$

$$f_x(x, y) = e^y - ye^x$$

$$f_y(x, y) = xe^y - e^x$$

Punti critici

$$\begin{cases} e^y \cdot ye^x = 0 \\ xe^y - e^x = 0 \end{cases} \quad \begin{cases} e^y = ye^x \\ x(ye^y) - e^x = 0 \end{cases} \quad \begin{cases} e^y = ye^x \\ (xy-1)e^x = 0 \end{cases} \quad \begin{cases} e^y = ye^x \\ xy-1=0 \end{cases} \quad \begin{cases} e^y = ye^x \\ x = \frac{1}{y} \end{cases} \quad \vee \quad \begin{cases} e^x = \frac{e^y}{x} \\ y = \frac{1}{x} \end{cases}$$

$$f_{xx}(x, y) = -ye^x$$

$$f_{xy}(x, y) = e^y - e^x$$

$$f_{yy}(x, y) = xe^y$$

$$H_H = \begin{bmatrix} -e & 0 \\ 0 & e \end{bmatrix} = -e^2 < 0 \quad \text{per cui } (1, 1) \text{ è un punto di sella}$$